

Project Finding Nemo

Search for the better search.

A phased strategy to rebuild trust in search, protect retention and create a controlled path from Apache Solr fundamentals to semantic and AI-assisted discovery.

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SEARCH STRATEGY RECOMMENDATION

SUPPORT PRESSURE

+32%

Year-on-year growth in search-related support tickets.

ACCOUNT EXPOSURE

7 / 20

Top accounts where search contributes to churn risk.

UNRESOLVED NEED

3 years

Of compounding user feedback without a focused response.

PORTFOLIO RECOMMENDATION

Fix the Solr foundation now, validate semantic retrieval beside it, and unlock generative experiences only after quality, trust and economics pass explicit gates.

Search is not an isolated feature problem. It is a trust and retention problem.

Users cannot reliably find the assets, campaigns and analytics they need, so the product feels less useful even when the information exists.

USER PROBLEM

People cannot predict what search will understand.

Inconsistent results, weak metadata and limited intent handling force users to reformulate, browse manually or ask support.

BUSINESS PROBLEM

Poor findability compounds cost and renewal risk.

Every failed search can become lost time, lower engagement, a support contact or fresh evidence in a difficult renewal conversation.

STRATEGIC PROBLEM

The comparison set has moved beyond direct competitors.

Customers now judge enterprise search against fast, conversational and AI-native experiences, raising the minimum bar for relevance and ease.

DECISION

Improve the core while earning the right to add AI.

The strategy must create near-term value, preserve governance and generate evidence for larger semantic or generative investment.

Turn three years of complaints into a measurable problem statement.

Use a quick survey to quantify the pattern, then interviews and behavioural data to explain why it occurs.

01

Instrument the search journey.

Track query, result set, first click, refinement, abandonment and downstream task completion as one connected funnel.

02

Segment the failure.

Compare roles, account tiers, content types and known-item versus exploratory tasks so averages do not hide the highest-value pain.

03

Translate friction into business exposure.

Join poor-search cohorts to support hours, product engagement, renewal health and revenue-at-risk rather than claiming direct revenue causality.

04

Sequence stakeholder discovery.

Learn from Customer Success, Support and Analytics first, validate constraints with Engineering and Design, then take evidence to Sales and executives.

BEFORE INTERVIEWS

Five-question search pulse survey

Quantify failure types, starting knowledge, success, workarounds and hardest tasks.

1. When search doesn't work the way you expect, what usually goes wrong?

SELECT ALL THAT APPLY

- | | |
|---|---|
| ☐ I get no results, even though I think the content exists | ☐ I get too many irrelevant results |
| ☐ The right result exists, but it appears too far down the list | ☐ I use different words from the words used to name/tag the content |
| ☐ The content doesn't have enough useful metadata/tags | ☐ I find multiple versions and don't know which one is current/correct |
| ☐ Content I expect to see doesn't appear because of access/permissions | ☐ My search is too specific or complicated for search to understand |
| ☐ I know roughly what I want, but not exactly what to search for | ☐ I'm looking for an answer/information rather than a specific asset |
| ☐ Search results take too long to load | ☐ Something else: _____ |
| ☐ I rarely have problems with search | |

This is the most valuable survey question because it maps responses directly to the search failure taxonomy.

2. When you're looking for something, what do you usually know about it?

SELECT ALL THAT APPLY

- | | |
|--|---|
| ☐ Exact filename or title | ☐ Part of the filename/title |
| ☐ Campaign/project name | ☐ Brand/product |
| ☐ Asset/content type | ☐ Date or approximate time period |
| ☐ Region/market | ☐ Person/team who created it |
| ☐ Tags or metadata | ☐ Words contained inside the document |
| ☐ What the asset looks like | ☐ I only know roughly what I'm looking for |
| ☐ Other: _____ | |

This separates known-item search problems from broader discovery problems.

3. How often do you successfully find what you're looking for using search alone?

SINGLE CHOICE

- | | |
|--|--|
| ☐ Almost always | ☐ Most of the time |
| ☐ About half the time | ☐ Occasionally |
| ☐ Rarely | ☐ Never |
| ☐ I don't use search enough to say | |

This creates a simple perceived-success baseline that can be measured again after improvements.

4. What do you normally do when search doesn't find what you need?

SELECT ALL THAT APPLY

- | | |
|---|--|
| ☐ Change the words in my search | ☐ Make the search shorter/simpler |
| ☐ Use filters | ☐ Browse through folders/categories |
| ☐ Search using another system/tool | ☐ Ask a colleague |
| ☐ Contact an administrator/support | ☐ Give up and use something else |
| ☐ Create/download another copy because I can't find the original | ☐ Other: _____ |

The workaround shows the true cost of failure; asking a colleague or recreating content signals much more pain than a general complaint.

5. Which type of search is hardest for you today?

SINGLE CHOICE

- | | |
|---|--|
| ☐ Finding a specific asset I know exists | ☐ Finding the latest/correct version of an asset |
| ☐ Finding something when I only remember part of its name | ☐ Finding something when I don't know how it was named/tagged |
| ☐ Finding assets based on what they contain | ☐ Browsing/discovering content when I don't know exactly what I want |
| ☐ Finding information or answers contained within documents | ☐ Filtering down a large number of results |
| ☐ Search generally works well for me | ☐ Other: _____ |

This helps distinguish lexical optimisation, semantic retrieval, versioning, metadata and filtering priorities.

Use qualitative interviews to explain the patterns behind the survey.

QUESTION 1

Tell me about the last time you needed to find an asset or piece of content. What were you looking for, and how did you go about finding it? Do you use any filters?

This reveals the user's real search workflow, where they begin, and whether they rely on keywords, browsing, filters, or another workaround.

QUESTION 2

What information do you usually know when you start searching?

For example: filename, campaign name, brand, date, asset type, person, tags, or just a rough idea of what you need. These details can become search prefixes or guided prompts, helping users start from what they already know and gradually adopt the terminology the system recognises.

QUESTION 3

Can you think of a recent example where you knew the right asset existed but still couldn't find it? What made it difficult?

This helps identify specific failure modes such as naming differences, missing metadata, weak ranking, unsuitable filters, or access constraints.

QUESTION 4

When you get search results, how do you decide which result is the right one?

This helps uncover important ranking signals like title, thumbnail, date, campaign, owner, approval status, etc.

QUESTION 5

Have you ever searched using synonyms or words that were different from those used to name or tag the asset? What happened?

This is a useful way to uncover a potential semantic-search problem without mentioning "semantic search."

Benchmark the outcomes customers value, not the features vendors advertise.

I would start with search-first products such as Amazon and sports-betting platforms, then examine familiar customer tools such as Canva and Google Drive to understand the search expectations they set before benchmarking direct competitors.

01 Relevance and findability

Can users find known items, recover from imperfect queries and see useful results in the first positions?

02 Refinement and control

Are facets, filters, permissions, saved searches and sorting clear enough for high-stakes enterprise work?

03 Semantic and conversational value

Does natural-language or semantic search solve tasks that keyword search misses, or only create an impressive demo?

04 Trust, speed and governance

Are results explainable, permission-safe, current, responsive and grounded in original source material?

The market-noise filter

A competitor feature earns roadmap attention only when it clears four tests.

1 Repeated customer need

Example: several customers report that they cannot find the latest approved asset.

2 Relevant segment and workflow

Example: marketing teams with large asset libraries face this problem every week.

3 Measurable outcome improvement

Example: the change reduces failed searches and time to a useful result.

4 Feasible and governable delivery

Example: Solr can support it without weakening permissions or exceeding speed and cost limits.

Anything else is a hypothesis to test, not a commitment to copy.

Direction A fixes today's search. Direction B extends what search can understand.

They carry different value, risk and timing, but share the same data, instrumentation and relevance foundations.

DIRECTION A

Strengthen the non-generative Solr experience.

Improve relevance, schema, metadata, facets, query handling and interface clarity in the system already powering search.

BEST FIT

Known-item lookup, exact fields, filtering, predictable enterprise workflows and permission-sensitive retrieval.

VALUE

Fastest route to broad improvements with lower delivery risk, explainable ranking and controlled operating cost.

TRADE-OFF

Still depends on query vocabulary and metadata quality, so ambiguous or conceptual searches may remain difficult.

RELATIVE PROFILE

Low-to-medium cost and risk; measurable value can begin in 6-12 weeks after baseline work.

DIRECTION B

Add semantic retrieval, then selective LLM experiences.

Use meaning-based retrieval for harder queries and add generated answers only where evidence shows they improve a real task.

BEST FIT

Exploration, vague intent, cross-content discovery and natural-language questions where users do not know exact terms.

VALUE

Recovers relevant items that lexical matching misses and can reduce the cognitive effort of complex discovery.

TRADE-OFF

Raises data, evaluation, latency, cost, privacy and hallucination concerns, especially for generated summaries.

RELATIVE PROFILE

Medium-to-high cost and risk; prototypes can be quick, but trustworthy production value usually takes 3-6+ months.

INITIAL INVESTMENT SPLIT

A / B
70 / 30

Fund both in sequence, with a small parallel discovery track.

Put most capacity into Solr foundations while a bounded team validates semantic value on the hardest query set; shift investment only when the prototype beats lexical search and meets security, latency and cost gates.

DEVELOPMENT

Prove feasibility and operating shape.

Run bounded spikes for schema, indexing, vector support, permissions, latency and cost before estimates become commitments.

DESIGN

Prove comprehension and workflow fit.

Prototype both directions against the same customer tasks and test refinement, explanations, confidence and recovery.

TOGETHER

Judge evidence through one scorecard.

Review real queries, technical traces and observed task outcomes together, then recommend the next investment gate.

HYBRID SEARCH IN SIMPLE LANGUAGE

Direction A builds the core; Direction B adds semantic reach.

Both paths meet inside Apache Solr. Direction A strengthens dependable search, while Direction B adds meaning-based signals and only later tests optional LLM answers.

DIRECTION A · 70% CORE INVESTMENT

Clean data, lexical relevance, filters, permissions, interface improvements and measurement.

DIRECTION B · 30% VALIDATION TRACK

AI enrichment, embeddings, dense-vector retrieval and later grounded-answer experiments.

The split represents team effort, not the number of steps. Some steps combine both directions inside the same Solr workflow.

1. PREPARE

DIRECTION A

Build reliable inputs

Normalise titles, text, metadata, permissions and freshness signals so Solr has dependable data.

2. ENRICH

DIRECTION B

Add meaning signals

AI suggests governed tags, entities and summaries; embeddings represent meaning as vectors.

3. RETRIEVE

A + B

Run both searches in Solr

A supplies keyword and facet retrieval. B adds dense-vector similarity under the same permissions.

4. BLEND

A + B

Combine the evidence

Controlled weights or reranking merge lexical and semantic results while preserving explainability.

5. PRESENT

A NOW · B LATER

Show trusted results

A presents original sources first. B can add a grounded answer only after retrieval proves trustworthy.

Important distinction

Semantic retrieval does not require a live generative LLM to write an answer. Embeddings represent meaning, while Solr retrieves and ranks the permitted source records.

IN-PRODUCT LEARNING LOOP

Invite users back, then make failed searches easy to explain.

Replace a high-effort support ticket with lightweight feedback captured at the moment of failure, and show users that their input has been logged for review.

1. INVITE

Try the improved search

Use a targeted in-product prompt to announce the change and bring users back to search after launch.

2. CAPTURE

Didn't find what you needed?

Carry across the query, filters and result state, then ask one optional question about what the user expected.

3. CLOSE THE LOOP

Confirm it is being reviewed

Acknowledge the submission, route it into the relevance backlog and offer an update when the issue is addressed.

Measure whether people complete the job, not whether they used the search box.

A shared scorecard links search quality to user behaviour, guardrails and business exposure.

01

Search task success

Percentage of search sessions that reach a defined downstream outcome without support or manual recovery.

EXAMPLE

A user searches for the Q3 campaign brief, opens the approved version and adds it to a campaign without asking Support.

02

Time to useful result

Median time from the first query to the first meaningful action, segmented by task and customer type.

EXAMPLE

A user submits a query and opens the correct approved asset 42 seconds later.

03

Failed-search rate

Zero-result, repeated reformulation and abandonment signals combined into a visible failure measure.

EXAMPLE

A search returns no results, the user tries two new queries, then leaves without opening an asset.

04

Relevance quality

How close the best answers are to the top, measured on searches judged by people who know the content.

EXAMPLE

For "latest APAC product launch video," the approved video appears within the top three judged results.

05

Business exposure

Search-related tickets, support hours, account-health movement and renewal-risk mentions over time.

EXAMPLE

Track whether accounts with high failed-search rates also create more tickets or mention findability during renewals.

Direction A proves reliability.

Expect lower zero-result and reformulation rates, better first-result relevance, faster task completion and fewer support contacts.

Direction B must prove incremental value.

Measure win rate on hard queries, grounded-answer accuracy, user preference, p95 latency and cost per successful task.

Response to the COO

This is not a promise of direct new revenue or two disconnected workstreams. It is a controlled investment in renewal protection, support efficiency and product adoption, with the semantic track funded only while it demonstrates incremental value over the repaired core.

Roll this out in stages, proving value before we invest more.

Each stage should improve the experience or answer a key question. We only move forward when the evidence is strong.

PHASE 0 | WEEKS 0-4

Understand the starting point

Measure search today, review real queries, speak with users and check Solr, metadata and access rules.

Move forward when: we can clearly explain who is struggling, with what tasks, and why.

PHASE 1 | WEEKS 5-12

Improve today's search

Tune ranking, improve indexing and query handling, make filters more useful and test changes with priority customers.

Move forward when: more users find what they need and fewer searches fail.

PHASE 2 | QUARTER 2

Test semantic search

Test meaning-based search on the hardest queries. Run it in the background first, then with a small customer group.

Move forward when: it finds better results without unacceptable speed, cost or privacy issues.

PHASE 3 | QUARTER 3+

Expand what works

Use keyword and semantic results more widely. Test AI answers only where they add clear value.

Move forward when: the improvement lasts, scales safely and earns user trust.

Search data and platform

Good metadata, readable content, a suitable Solr version, enough capacity to update the index and clear monitoring.

Security and privacy

Search must respect access rights, keep each customer's data separate and meet privacy, location and audit rules.

User adoption and feedback

Test the experience, explain what changed, prepare Customer Success and collect feedback inside the product.

AI quality, speed and cost

Check answer quality, model changes, security risks, response time and running cost.

Customer and commercial timing

Plan around renewals, customer promises, contract limits and accounts where poor search creates the most risk.

Ongoing search quality

Assign clear ownership for test queries, experiments, quality checks and regular ranking improvements.

Alignment comes from shared evidence, explicit decisions and predictable cadence.

The product leader owns the frame, the decision process and the connection between customer outcomes and investment.

WEEKLY

Delivery working group

Product, Engineering, Design, Data and Support review evidence, blockers, quality and experiment outcomes.

MONTHLY

Executive decision review

Stakeholders review scorecard movement, risk and investment gates.

QUARTERLY

Customer advisory check

Priority accounts validate whether roadmap progress changes real workflows, trust and renewal confidence.

When senior stakeholders disagree

1. Make the contested assumption explicit: customer value, feasibility, cost, risk or timing.
EXAMPLE: Sales believes semantic search will protect renewals, while Engineering believes poor metadata is the real constraint. Frame the question clearly: which change improves search success most for at-risk accounts?
2. Use the pre-agreed scorecard and decision gates, not role seniority, as the default arbitration mechanism.
EXAMPLE: Compare the Solr and semantic options against the same measures: task success, relevance, speed, cost and permission safety.
3. Run the smallest time-boxed experiment that can resolve the disagreement and record the decision.
EXAMPLE: Test standard keyword search and hybrid search against 100 difficult real-world queries for two weeks before funding a production build, then record the result and decision.
4. Commit to the chosen path until the next review trigger, then update openly when evidence changes.
EXAMPLE: Keep the 70/30 investment split for one quarter and revisit it only at the agreed review if semantic search beats the core search on quality, speed and cost.

More detail, explained with practical examples.

Open these sections when someone wants to understand the reasoning, evidence or delivery approach behind the main presentation.

A1

Why the 70/30 recommendation makes sense

The recommendation improves the search customers use today while testing a more advanced option on a smaller scale. It avoids putting the whole investment into either the current approach or an unproven AI experience.

Why not choose only Direction A?

Improving Solr should fix many common search problems, but keyword search will still struggle when users do not know the right name or terminology.

EXAMPLE: A user searches for “family beach campaign,” but the asset is named “Coastal Summer Launch.” Better keyword search may still miss the connection.

Why not start with Direction B?

Meaning-based search and AI answers still depend on good data, correct permissions and reliable evaluation. Adding AI first could make existing problems harder to understand.

EXAMPLE: If an asset has the wrong access settings, an AI answer could expose information the user should not see.

What we need to confirm

- Most searches can improve through better ranking, metadata, filters and interface design. **Example:** finding an approved logo by brand and region.
- A smaller group of valuable searches depends on meaning rather than exact words. **Example:** finding “content that feels adventurous” when no asset uses that phrase.
- The current Solr setup can support meaning-based vector search, or can be upgraded without a full rebuild.
- The same access rules can be applied to both keyword and meaning-based results.
- Search failures can be connected to support effort, lower product use or renewal concern.
- For important work, customers prefer a result they can verify over an impressive answer with no visible source.

Product data tells us how often search fails. Surveys and interviews tell us why it fails and what users do next. We need both before choosing a solution.

Weeks 1-2: understand the data

- Check what search activity is already tracked.
- Review searches with no results, repeated wording changes or no result opened.
- Identify the customers, roles and tasks most affected.
- Create a set of real searches that we can use to judge improvements.

EXAMPLE: Review 100 recent failed searches and group them into problems such as missing metadata, poor ranking or different terminology.

Weeks 2-3: learn from users

- Interview 8-12 users from different customer types.
- Watch them find a known asset and explore when they are less certain.
- Compare what users say with what they actually do.
- Record workarounds and what makes a result feel trustworthy.

EXAMPLE: A user says search is "fine," but during the task they browse folders and ask a colleague instead.

Weeks 3-4: agree what to solve

- Rank problems by frequency, customer impact and business risk.
- Check possible solutions with Engineering and Design.
- Agree how success will be measured and when to stop.
- Present the evidence and recommendation to leaders.

EXAMPLE: Prioritise poor ranking for approved campaign assets if it affects many renewal-sensitive accounts.

EVIDENCE	WHAT IT TELLS US	ILLUSTRATIVE EXAMPLE
Support tickets and handling time	Which problems keep repeating and how much service effort they create.	40 search tickets taking 30 minutes each represent about 20 hours of support effort.
Behaviour after a failed search	Whether users keep trying, leave, browse folders or contact Support.	Compare accounts with frequent failed searches against similar accounts where search succeeds.
Renewal notes and risk flags	Where poor search is reducing confidence in the product.	If two at-risk accounts mention findability, report their contract value as "revenue at risk," not revenue already lost.
Time spent and workarounds	The effort hidden outside normal product analytics.	If finding an asset takes five extra minutes and happens 200 times a week, the lost time becomes visible.

Who to involve, and when

1. Start with Customer Success, Support and Analytics to establish what is happening.
2. Speak with customers and Design to understand why.
3. Bring in Engineering, Data and Security to test what is possible and safe.
4. Use Sales and Renewal teams to confirm business timing and risk.
5. Ask executives to decide scope once the evidence is visible.

EXAMPLE: Do not begin with "Should we buy an AI search tool?" Begin with "Which customer tasks are failing, and why?"

How to explain the business impact

Use a sensible range instead of claiming exact revenue. Show where search is a documented contributor to risk, support cost and reduced product use.

EXAMPLE: "Search is mentioned in three renewal-risk accounts worth \$X-\$Y, and creates approximately Z support hours each month."

How I would review the market

I would start with products where search is a main way to navigate, such as Amazon and sports-betting platforms. I would then review familiar tools such as Canva and Google Drive because they shape what customers expect, before comparing direct competitors and AI search products.

Use the same tasks in every product

- Find a known asset when only part of the name is remembered.
- Narrow results by campaign, date, region and usage rights.
- Find related content using an everyday description.
- Ask a question across several documents and check the sources.
- Try to find restricted content and see whether access rules hold.

EXAMPLE: Search every product for “latest approved APAC launch video” and record how many steps it takes to reach the correct item.

Score usefulness, not visual polish

- Did the user complete the task?
- Was the correct result near the top?
- Could the user recover from a weak or incomplete search?
- Were filters, explanations and sources clear?
- Were results fast and permission-safe?

EXAMPLE: A conversational answer should not score well if the user cannot verify its source or open the original asset.

AREA	QUESTION TO ASK	EXAMPLE OF USEFUL EVIDENCE
Findability	Does this help with an important task customers perform often?	Users find the approved asset faster and need fewer search attempts.
AI capability	Does AI solve a real problem, or only make the interface look newer?	Meaning-based search finds relevant assets that improved keyword search still misses.
Trust and control	Can users check, refine and safely use the result?	The product shows sources, respects permissions and lets the user adjust filters.
Customer value	Would target customers care enough to use it?	Priority customers ask to join a beta and use the feature repeatedly during testing.

How Direction A and Direction B work together

Direction A improves the dependable search experience customers use today. Direction B adds meaning-based search for harder tasks. They use the same data, permissions and Solr platform.

DIMENSION	DIRECTION A: IMPROVE SOLR SEARCH	DIRECTION B: MEANING-BASED SEARCH AND OPTIONAL AI ANSWERS
What changes	Improve keyword ranking, spelling, synonyms, filters, metadata and result design.	Add meaning-based matching and, later, optional answers supported by source content.
Best for	Users who know a filename, brand, date, asset type or filter.	Users who only know the idea, intent or information they need.
Example	"2026 APAC launch brochure" matches the title, date and region fields.	"Family-friendly summer content" finds relevant assets even when those exact words are not used.
Main benefit	Improves a large share of searches quickly and is easier to explain.	Finds useful content when the user's words differ from the asset's words.
Main trade-off	Still depends on good metadata and recognisable words.	Needs more testing for quality, privacy, speed, cost and trust.
How we prove it	More successful searches, fewer failed searches and faster task completion.	Better results on difficult queries than Direction A alone, with safe permissions and acceptable cost.
Starting investment	70% while the core search is being repaired.	30% to build and test a small prototype before expanding.

How meaning-based search works

1. Collect the searchable text and approved metadata.
2. Use an embedding model to turn the meaning into a list of numbers called a vector.
3. Store that vector in Solr beside the normal keyword fields and access rules.
4. Turn the user's search into a vector in the same way.
5. Let Solr compare both keyword matches and meaning matches, then return permitted source records.

EXAMPLE: "Coastal Summer Launch" and "family beach campaign" use different words, but their vectors may be close because their meanings are related.

How AI can improve the data

- Suggest missing tags, themes, people, regions or short summaries.
- Keep customer-entered metadata as the official source and clearly label AI suggestions.
- Record which model created each suggestion and how confident it was.
- Send uncertain or sensitive suggestions for human review.
- Only repeat the process when it creates enough search improvement to justify the cost.

EXAMPLE: AI may suggest "outdoor," "family" and "summer" for an image, while an administrator approves or rejects those tags.

What must be true for A

A meaningful share of failures comes from ranking, metadata, filters or result design, and the current Solr setup can be improved without rebuilding everything.

EXAMPLE: The correct asset already appears, but sits on page three because recent approved content is not ranked highly enough.

What must be true for B

Some valuable tasks depend on meaning rather than exact words, and meaning-based search can beat the improved keyword baseline at an acceptable cost.

EXAMPLE: Users describe the visual they need but do not know its campaign name or tags.

How Design and Engineering work together

Engineering checks whether it can be built safely and reliably. Design checks whether it helps people complete real tasks. Both test the same searches.

EXAMPLE: Engineering measures response time while Design observes whether users understand and trust the reordered results.

Solr remains in control

Solr continues to apply permissions, filters, freshness and ranking rules.

EXAMPLE: A user only receives assets they are already allowed to open.

AI answers are optional

An LLM may summarise approved search results for a proven use case, but it does not replace the original sources.

EXAMPLE: An answer summarises three campaign briefs and links to each one.

Users must be able to verify

Show sources, explain uncertainty and provide a clear path back to standard results.

EXAMPLE: If confidence is low, show the most relevant documents instead of inventing an answer.

How we measure whether search improved

Use a small set of measures that show whether users completed the job, how much effort it took and whether the change reduced business risk.

METRIC	PLAIN-LANGUAGE MEANING	EXAMPLE
Search task success	The percentage of search sessions that end in the intended action.	If 80 of 100 users find and open the approved asset, task success is 80%.
Time to useful result	The typical time from the first search to the first useful action.	The user opens the correct campaign brief 42 seconds after their first search.
Failed-search rate	The share of searches with no results, repeated rewrites or no useful action.	The user tries three versions of a search and leaves without opening anything.
Ranking quality	How close the best answers are to the top, measured on searches judged by people who know the content.	For “latest APAC launch video,” the approved video appears in the top three results.
Business exposure	Support effort and customer risk connected to search problems.	Accounts with frequent failed searches also raise more tickets or mention findability during renewals.

Measure the starting point

1. Choose the 3-5 most important search tasks.
2. Define what success looks like for each task.
3. Collect at least four weeks of reliable data.
4. Build a reusable set of real searches from logs and interviews.
5. Ask more than one expert to judge the expected results.

EXAMPLE: For “find the latest approved asset,” success means opening the current approved version, not simply clicking any result.

Check whether it improved

1. Test ranking changes against the saved query set before customers see them.
2. Run meaning-based search in the background first and compare its results.
3. Show different versions to small user groups where traffic allows.
4. Combine product data with observed customer tasks.
5. Check each customer segment so an overall average does not hide a worse experience.

EXAMPLE: A change may improve results for large libraries but make small libraries slower; report both rather than only the combined average.

Example conditions for moving forward

- Direction A: task success improves without making search slower or weakening access rules. **Example:** success rises from 65% to 75%.
- AI answers: every answer links to its sources and no restricted information is exposed.
- Direction B: meaning-based search clearly beats Direction A on the difficult-query set. **Example:** it wins 30 of 50 searches that keyword search misses.
- Cost: the agreed cost per successful search remains affordable at expected usage.

Start small, watch what happens and make it easy to return to the existing search. Expand only when quality, safety, speed and cost remain within agreed limits.

Start with a small rollout

- Test in the background and with internal users first.
- Turn the new experience on for selected accounts or content types.
- Prompt users to try it and keep a clear path back to standard search.
- Add a “Didn't find it?” action that keeps the query, filters and result state.
- Confirm that feedback was received and send it to an owned improvement backlog.
- Agree in advance what would cause us to pause or roll back.

EXAMPLE: Begin with five volunteer accounts and image assets, then expand only if search success improves and access rules remain correct.

Keep search quality stable

- Keep a record of every important data, model and ranking change.
- Retest saved queries before releasing each change.
- Keep AI-suggested metadata separate from customer-authored metadata.
- Check whether model quality or stored vectors become outdated.
- Treat search improvement as ongoing work, not a one-time project.

EXAMPLE: Before changing synonym rules, rerun the saved searches to confirm that “brand guide” still returns the approved document first.

Protect trust, speed and cost

- Apply access rules before showing results or generating an answer.
- Use approved models with clear data-retention terms.
- Reuse stored results only when permissions remain correct.
- Track slow searches and the cost of each successful task.
- Do not expand an AI feature that cannot show its sources.

EXAMPLE: If an AI answer takes eight seconds or cannot cite the source document, show normal search results instead.

RISK	WHAT COULD HAPPEN	EXAMPLE RESPONSE
Poor or outdated metadata	Keyword and meaning-based results both become noisy.	If old campaign dates push expired assets upward, fix those fields and re-index the affected content.
Access leak	A user sees a result or summary from content they cannot open.	Apply account and permission filters before ranking or generation, and stop rollout after any confirmed leak.

RISK	WHAT COULD HAPPEN	EXAMPLE RESPONSE
Made-up AI answer	An answer invents information or describes a source incorrectly.	Require source links, restrict answers to retrieved content and show documents instead when confidence is low.
Slow or expensive search	Better results take too long or cost too much to run at scale.	Set speed and cost limits, choose smaller models where possible and disable features that exceed them.
User confusion	People distrust changed rankings or an unfamiliar answer experience.	Introduce changes gradually, explain why results appear and provide an easy feedback option.

Who owns what and how decisions are made

Each group contributes different evidence. Product brings it together into one recommendation, while agreed review points determine whether investment changes.

GROUP	WHAT THEY OWN	EXAMPLE DECISION OR CONTRIBUTION
Product	The problem, priorities, success measures and recommendation.	Recommend the 70/30 split and show which evidence would justify changing it.
Engineering and Data	The Solr design, tracking, reliability, estimates and running cost.	Confirm whether vector search can meet the required response time and permission rules.
Design and Research	User tasks, result refinement, explanations and trust.	Observe whether users understand why a result appears and can recover from a poor search.
Customer Success, Support and Sales	Customer context, research recruitment, adoption and renewal signals.	Recruit priority accounts for a beta and identify customers where search affects confidence.
Security, Legal and Finance	Privacy, data use, supplier terms, risk and affordability.	Approve how model providers may store data and set the acceptable cost range.
Executive sponsor	Cross-team priorities and major investment changes.	Protect the agreed team capacity and approve expansion only when the review conditions are met.

Tell one consistent story

The initiative improves customer value now and tests a better future option. The emphasis may change by audience, but the evidence and success measures stay the same.

EXAMPLE: Finance hears about cost and renewal risk; Engineering hears about reliability and sequencing; both see the same task-success score.

Keep a simple decision record

Record the question, evidence, owner, decision, concerns and date for review. Reopen the decision only when new evidence appears or the review date arrives.

EXAMPLE: "Keep 70/30 for Quarter 2. Review after the 100-query test. Product owns the decision; Engineering owns the performance evidence."

DECISION SUMMARY

Fix what is broken. Prove what is possible. Scale only what earns trust.

Keep Apache Solr as the governed retrieval core, improve the deterministic experience first, validate semantic lift on real customer tasks and reserve generative AI for grounded use cases where it adds measurable value.

PERMISSION NOTICE

This strategy is provided for assessment and discussion only. The implementation concepts outlined here may not be used or implemented without prior written permission.

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